



General Format

This Mounting Mayhem round is about a type of puzzle called **Cave**. The round consists of some classic Cave puzzles, as well as some variants that slightly modify the rules of the puzzle. You will have 15 minutes to read through this rules document, and then 60 minutes to complete the puzzles. The puzzles are divided into 4 variants:

- **Classic Cave:** 7 puzzles
- **Variant 1 (Product):** 3 puzzles
- **Variant 2 (No 2x2):** 3 puzzles
- **Variant 3 (Both):** 2 puzzles

Each puzzle will have an approximate difficulty rating from 1 to 5 stars. No partial credit will be given to incorrect puzzles, even if the errors are minor. It is highly recommended that all members of your team read all of the provided rules, and perhaps discuss strategies if time allows.

Submission Instructions

Your team will receive 4 “scratch paper” packets of the puzzles that you can work on to solve the puzzles in the round. You may take these apart, give one to each team member, etc.

Your team will also receive one final answer packet, which will contain a copy of each puzzle from the round — you should NOT take this apart. In addition, you should only draw on this sheet once you are confident in your answer. Shade the grid clearly, and if you have to erase, erase neatly! Smudged or confusing boxes may result in incorrect grading.

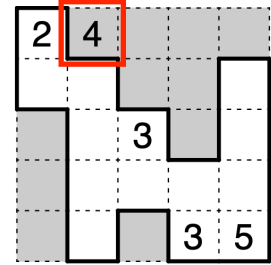
Each puzzle on the answer sheet will have a number of points, as well as a difficulty rating associated with it. You can use this information to inform your strategies in this round. **We highly discourage attempting the puzzles from top to bottom!** The hardest puzzles in one section will be more difficult than the easiest puzzles in the next section.

1 Introduction to Caves

A standard Cave is played on a rectangular grid, with numbers in the some cells formed by this grid. Your goal is to shade certain squares such that the resulting “cave” (a region of unshaded cells) follows some rules. As you read through the rules, try to cross-reference with the example on the right which **breaks the rule (indicated in red)** to make sure you understand! (Note: unless otherwise stated, these examples are valid solutions *other* than breaking the rule to the left of said example.)

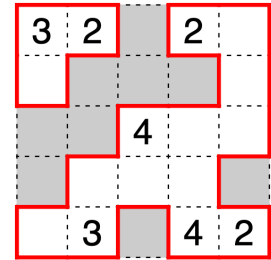
- [N] **Number Rule**

Numbers cannot be shaded.



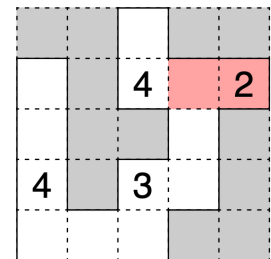
- [S] **Singularity Rule**

All unshaded cells must be connected to each other orthogonally (via edges, not corners). That is, there can only be 1 **continuous cave** of unshaded cells in the grid.



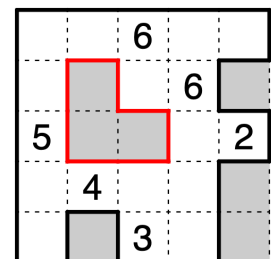
- [G] **Given Rule**

A number clue within a cell indicates the number of unshaded cells in a straight line horizontally or vertically that it can “see,” including itself. A cell cannot “see” any shaded cells or any cells where a shaded cell blocks its view.



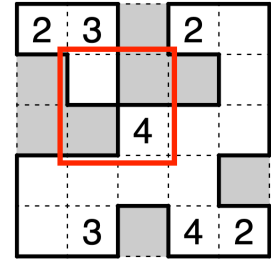
- [NS] **No Stalagmite Rule**

All shaded cells should connect via other cells to the **borders** of the grid. So, no shaded cell should be surrounded by unshaded cells in all four orthogonally connected cells.



- [AC] **Avoid a Checkerboard**

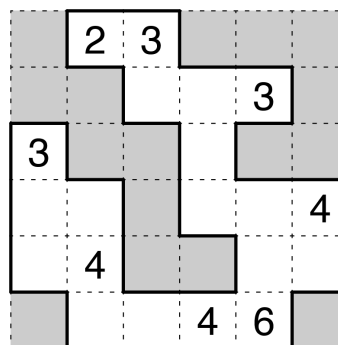
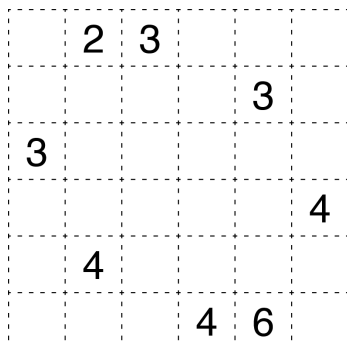
No **2x2** region of the grid should be shaded like a checkerboard — with two unshaded cells and two shaded cells touching only at one corner. (Note that this results from the singularity [S] rule.)



As you solve, it is often helpful to mark the cells with different colors or textures to help you keep track of which cells are shaded or unshaded. **Please DO NOT mark unshaded cells on the final submission sheets.**

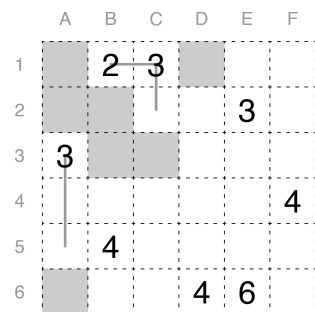
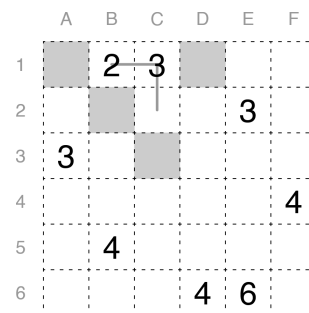
All puzzles in the Mounting Mayhem Round have unique solutions that can be deduced logically. If you believe that there are multiple solutions, it is likely that you have likely misinterpreted the rules of the puzzle.

Example Walkthrough



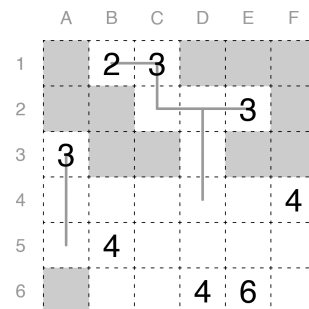
Let's walk through the solving process for the Cave above. **[Bolded letters]** indicate common patterns that appear throughout the walkthrough (see below). Additionally, we will number the columns from A through F going left to right, and we will number the rows from 1 through 6 going top to bottom.

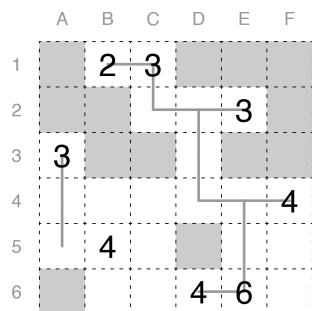
It is important to remember that all numbers are unshaded **[N]**. Because of this, the 2 clue in the upper left already sees two cells. This means that it cannot see any more cells horizontally or vertically, leading to three shaded cells **[DE]**. Furthermore, the 3 clue must see exactly one more cell vertically, leading to the progress shown.



At this point, there are two ways we can progress. First, recall that we must not form a checkerboard pattern of shaded and unshaded cells **[AC]**. Cell A2 must be shaded to avoid this pattern. Similarly, cell B3 must be shaded to avoid a checkerboard. These developments allow us to completely determine the 3 clue in the third row **[OD]**.

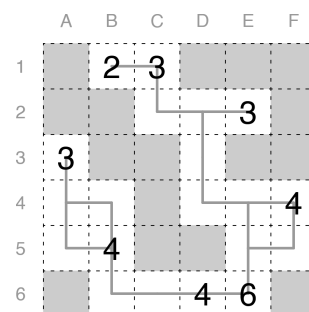
Let's now think about connectivity **[S]**. How can the white cells in the top row escape? The only way is through cell D2; this cell must be unshaded. From here, the 3 clue in cell E2 is complete, so we mark the remaining cells around it as shaded. Remember not to make a checkerboard **[AC]**. Furthermore, the white cells must escape into row 4 **[S]**.





Now focus on the 4 clue in cell D6. The cell right above it cannot be unshaded, or else this 4 clue would see 5 unshaded cells above it [DE]. By itself, this does not do much. However, compare the 4 clue with its partner 6 [AXY]. These two clues see the same number of cells horizontally. It follows that the 6 must see two more cells vertically.

Finally, note that the 4 clue must take at least one cell to its left because it sees exactly 4 cells horizontally [OD] and there are only three cells to its right [LC]. All shaded cells must escape to the edges of the grid [NS]. We have an isolated shaded cell that must escape through the top left. From here, the 4 clue in cell B5 is complete [LC], and we may finish the puzzle.

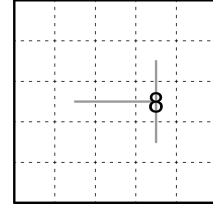


Common Patterns

Some steps are common in Cave puzzles. Let's expand on a few of them here.

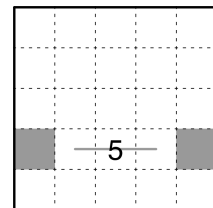
- **[LC] Large Clues**

If a clue is large enough, you can check which cells it must be able to see. In a 5×5 cell, any 9 clue should see all the cells horizontally and vertically, and a 8 clue can see all but one. This one clue must be one of the four border cells in the 8's line, so you can mark any cells between the 8 and the border cell as unshaded.



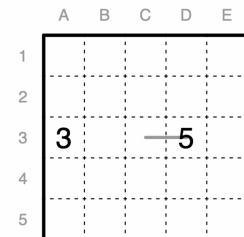
- **[OD] One Direction Remaining**

If a clue cannot see any more cells horizontally, then we know exactly how many cells it must see vertically. For example, in the diagram to the right, the 5 clue sees 3 cells horizontally, so it *must* see $5 - 3 + 1 = 3$ cells vertically. (Remember, the clue itself is counted only once here!)



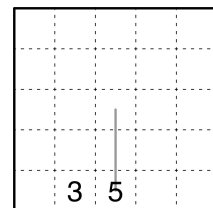
- **[DE] Don't Extend Too Far**

Each clue must see *exactly* the number of cells given, not more or less. Sometimes, this means that cells must be shaded to "block" the line of sight of a particular clue. In the grid to the right, the 5 and the cell to its left (C3) are known to be unshaded. Then, the cell in B3 must be shaded. Otherwise, the 3 clue to its left would see at least 4 unshaded cells.



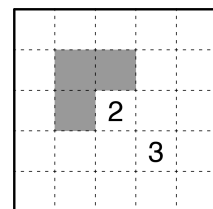
- **[AXY] Adjacent Clues**

Suppose X and Y are adjacent number clues with $Y \geq X$. They see all the unshaded cells in the same line (horizontal or vertical). In the other direction (vertical/horizontal), the larger clue Y must see $Y - X$ more cells than X . For example, if a 3 and 5 are adjacent to each other horizontally, then the 5 must "see" $5 - 3 = 2$ more cells vertically.



- **[D2X] Diagonal Two Clue**

Here's a trickier deduction. When a 2 is diagonal to an unshaded cell X , the two cells that the 2 sees must include itself and one of the cells adjacent to both the 2 and the X . (Otherwise, both shaded cells would form a checkerboard!) So, any cells adjacent to the 2 but not to the X must be shaded.

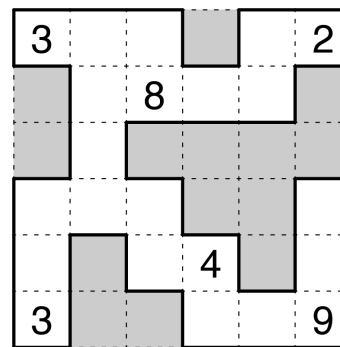
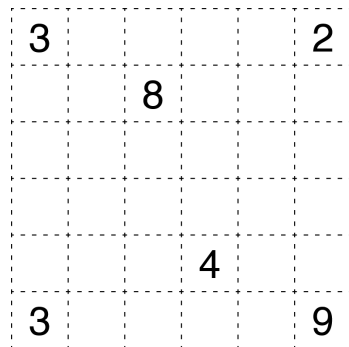


2 Product Cave

In this variant, most cave rules stay the same. The only difference is that number clues are defined differently.

- **[MG] Multiplicative Given Rule**

A clue indicates the product of the number of cells it sees horizontally (including itself) and the number of cells it sees vertically (including itself).

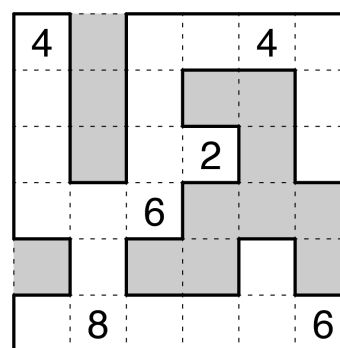
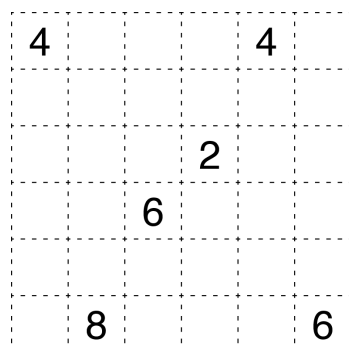


3 No 2×2

The rules of *classic* Cave stay the same – do not confuse this with multiplicative caves. Here, we have an extra condition the solution has to satisfy.

- **[2X2] 2×2 Shading Rule**

No 2×2 section of the cave can be entirely shaded or entirely unshaded.



(Note that the right-hand side of this puzzle cannot possibly be unique as a Classic Cave!)

4 Product + No 2×2 Cave

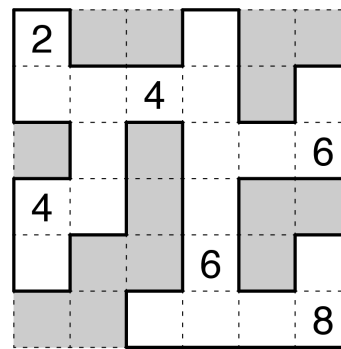
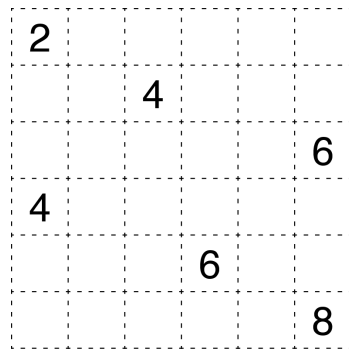
This is a combination of the Multiplicative Cave and No 2×2 Cave variants. To reiterate:

- **[MG] Multiplicative Given Rule**

A clue indicates the product of the number of cells it sees horizontally (including itself) and the number of cells it sees vertically (including itself).

- **[2X2] 2×2 Shading Rule**

No 2×2 section of the cave can be entirely shaded or entirely unshaded.



Wrapup

You should be all set for the Mounting Mayhem Round now! If the reading period is not over yet, consider rereading key sections or discussing strategies with your team. Good luck!